Proofs for series tests

Week 5

Definition

$$\text{Recall } \left(a_k\right) = \left(a_1, a_2, a_3, \ldots\right) \text{ is a sequence.}$$

$$\left(s_n\right) = \left(a_1, a_1 + a_2, a_1 + a_2 + a_3, \ldots\right), \text{ or } s_n = \sum_{k=1}^{n} a_k \, .$$

$$\lim\left(s_n\right) = \sum_{k=1}^{\infty} a_k \text{ is a series.}$$

$$\left(s_n\right) \text{ converges iff } \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Theorem 3.3. Geometric Series

A series is called geometric if it is of the form

$$\sum_{k=1}^{\infty} ar^{k-1}\,, \text{ with } \; r, a \in \mathbb{R} \setminus \{0\}\,.$$

Then a geometric series is convergent

if and only if $|r| < 1$.

Proof. Suppose $r \neq 1$.

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}$$

$$(-) \quad rs_n = ar + ar^2 + ar^3 \cdots + ar^n$$

$$(1-r)s_n = a(1-r^n) \Rightarrow s_n = \frac{a(1-r^n)}{1-r}, r \neq 1$$

From Worksheet 3, Q14

$$\lim\left(x^n\right) = 0 \text{ if } \left|x\right| < 1.$$

Hence, $\sum_{k=1}^{\infty} ar^{k-1}$ converges (to $\frac{a}{1-r}$) if $\left|r\right| < 1$, $r \neq 0$.

If $\left|r\right| > 1$, then $\lim\left(ar^{k-1}\right) = \pm\infty$.

If $r = 1$, then $\lim\left(ar^{k-1}\right) = a \neq 0$.

So, by the n-th term test (divergence test),

$\sum_{k=1}^{\infty} ar^{k-1}$ diverges if $\left|r\right| \geq 1$.

Hence, a geometric series is convergent if and only if $|r| < 1$.

Theorem 3.5

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be convergent series with $\sum_{k=1}^{\infty} a_k = \alpha$ and $\sum_{k=1}^{\infty} b_k = \beta$. Then

(a) $\sum_{k=1}^{\infty} \left(c a_k \right) = c\alpha$ and

(b) $\sum_{k=1}^{\infty} \left(a_k + b_k \right) = \alpha + \beta$,

where $c, \alpha, \beta \in \mathbb{R}$ are constants.

Proof. (a) Let $s_n = \sum_{k=1}^{n} a_k$.

Given $\sum_{k=1}^{\infty} a_k = \alpha$, meaning that $\lim\left(s_n\right) = \alpha$.

Applying the Algebraic Limit Theorem for sequences (Theorem 6.1, Chapter 2) yields

$$\lim\left(cs_n\right) = \sum_{k=1}^{\infty} ca_k = c\alpha \text{ , as desired.}$$

(b) Let $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} b_k$. Given that $\left(s_n\right)$, $\left(t_n\right)$ converge to α and β, respectively.

By the Theorem 6.1 of Chapter 2,

$\left(s_n + t_n\right)$ converge to $\alpha + \beta$.

Hence,

$$\lim\left(s_n + t_n\right) = \sum_{k=1}^{\infty}\left(a_k + b_k\right) = \alpha + \beta .$$

Recall: Cauchy sequence

Let $\left(x_r\right)$ be a sequence. Then $\left(x_r\right)$ is said to be Cauchy if

$$\boxed{\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > m > N \Rightarrow \left|x_m - x_n\right| < \varepsilon \ \forall n, m \in \mathbb{N}.}$$

Theorem: $\left(x_r\right)$ is Cauchy iff $\left(x_r\right)$ is convergent.

Theorem 4.1. Cauchy Criterion for Series

$$\boxed{\sum_{r=1}^{\infty} a_r \quad \text{converges}} \iff \boxed{\begin{aligned} &\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > m > N \Rightarrow \\ &\left| a_{m+1} + a_{m+2} + \cdots + a_n \right| < \varepsilon \ \forall n, m \in \mathbb{N} \end{aligned}}$$

Proof. Let $s_k = \sum_{r=1}^{k} a_r$. If $n > m$, then $s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n$.

$\Rightarrow$) Suppose $\left(s_n\right)$ converges. Then $\left(s_n\right)$ is Cauchy and it is done.

Because $\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \ni n > m > N \Rightarrow$

$$\left|a_{m+1} + a_{m+2} + \cdots + a_n\right| = \left|s_n - s_m\right| < \varepsilon \; \forall n, m \in \mathbb{N} .$$

$\Leftarrow$) Suppose $\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \ni n > m > N \Rightarrow$

$$\left|s_n - s_m\right| = \left|a_{m+1} + a_{m+2} + \cdots + a_n\right| < \varepsilon \; \forall n, m \in \mathbb{N} .$$

Then $\left(s_n\right)$ is Cauchy and so $\left(s_n\right)$ is convergent and it is done.

Alternative: Merging two in one

Proof. Let $s_n = \sum_{k=1}^{n} a_k$. Note that

$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n$. Then

$$\sum_{k=1}^{n} a_k \text{ converges}$$
$$\Leftrightarrow \left(s_n\right) \text{ converges}$$
$$\Leftrightarrow \left(s_n\right) \text{ is Cauchy}$$
$$\Leftrightarrow \forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > m > N \Rightarrow \left|s_n - s_m\right| < \varepsilon$$
$$\Leftrightarrow \forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > m > N \Rightarrow \left|a_{m+1} + a_{m+2} + \cdots + a_n\right| < \varepsilon$$

Theorem 4.1 is used to prove...

Theorem 4.1
(Cauchy Criterion for Series)

↗

Theorem 4.2
(n-th term test, aka divergence test)

↘

Theorem 4.7
(comparison test)

Theorem 4.2. (*n*-th term test, aka divergence test)

Theorem 4.2. $\sum_{k=1}^{\infty} a_k$ converges $\Rightarrow \lim\left(a_r\right) = 0$.

Proof. Take $\Rightarrow$) of Theorem 4.1 and let $n-1 = m < n$. Then

$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n, m > N \Rightarrow$

$$\left|s_n - s_{n-1}\right| = \left|a_n - 0\right| < \varepsilon \ \forall n > N, n \in \mathbb{N}.$$

Hence, by the definition of convergence sequence, $\lim\left(a_r\right) = 0$.

Remark

Remark: 1. Here we apply the $\Rightarrow$) direction of Theorem 4.1.

2. When $\sum_{k=1}^{\infty} a_k$ converges, it implies that $\left|a_{m+1} + a_{m+2} + \cdots + a_n\right| < \varepsilon$ for <u>all</u> $n > m$. So this result is true for <u>one case</u> of n .

3. Theorem 4.2 does not apply the $\Leftarrow$) direction of Theorem 4.1 because one case of n is correct does not implies that all cases of n is true.

Theorem 4.7. Comparison Test

Let $\left(a_k\right)$ and $\left(b_k\right)$ be sequences such that $0 \leq a_k \leq b_k \ \forall k \in \mathbb{N}$.

(i) If $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

(ii) If $\sum_{k=1}^{\infty} a_k$ diverges, then $\sum_{k=1}^{\infty} b_k$ diverges.

Proof. (i) Suppose $\sum_{k=1}^{\infty} b_k$ converges. Then by Theorem 4.1,

we have $\forall \varepsilon > 0 \; \exists N \in \mathbb{N}$ such that $n > m > N$ implies that

$$\left| b_{m+1} + b_{m+2} + \cdots + b_n \right| < \varepsilon \; \forall n, m \in \mathbb{N} .$$

$$\begin{array}{ccc} \backslash\!\vee & \backslash\!\vee & \backslash\!\vee \\ a_{m+1} & a_{m+2} & a_n \end{array}$$

Since $0 \leq a_k \leq b_k \; \forall k \in \mathbb{N}$, it follows that

$$\left| a_{m+1} + a_{m+2} + \cdots + a_n \right| \leq \left| b_{m+1} + b_{m+2} + \cdots + b_n \right| < \varepsilon \; \forall n, m \in \mathbb{N} .$$

Thus, by Theorem 4.1, $\sum_{k=1}^{\infty} a_k$ converges.

(ii) This is an immediate result from contrapositive of (i).

Example 4.4. Harmonic series

Prove that the Harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent.

Proof. Let $s_{2^n} = \sum_{k=1}^{2^n} \frac{1}{k}$ be a partial sum. Then

$$
\begin{aligned}
s_{2^n} &= \sum_{k=1}^{2^n} \frac{1}{k} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{n-1}+1} + \frac{1}{2^{n-1}+2} + \cdots + \frac{1}{2^n}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^n} + \frac{1}{2^n} + \cdots + \frac{1}{2^n}\right) \\
&= 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) + 2^2\left(\frac{1}{8}\right) + \cdots + 2^{n-1}\left(\frac{1}{2^n}\right) \\
&= 1 + \underbrace{\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2}}_{n \text{ times}} \\
&= 1 + \frac{n}{2}
\end{aligned}
$$

$$s_{2^n} = \sum_{k=1}^{2^n} \frac{1}{k} > 1 + \frac{n}{2}$$

$$\Rightarrow \lim\left(s_{2^n}\right) = \sum_{k=1}^{\infty} \frac{1}{k} > \lim\left(1 + \frac{n}{2}\right) = \infty$$

Therefore, $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent.

Theorem 4.5. (Cauchy Condensation Test)

Let $\left(a_k\right)$ be a monotone decreasing sequence with $a_k \geq 0 \ \forall k \in \mathbb{N}$. Then,

$$\sum_{k=1}^{\infty} a_k \text{ converges if and only if } \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges.}$$

In other words, if we have two sequences,

This is decreasing →

$$\left(a_n\right) = \left(a_1, a_2, \ldots, a_n, \ldots\right)$$

$$\left(b_n\right) = \left(2^{n-1} a_{2^{n-1}}\right) = \left(a_1, 2a_2, 4a_4, 8a_8, 16a_{16}, \ldots, 2^n a_{2^n},\right)$$

Start from 0, not from 1
So we write n – 1.

$$\boxed{\sum_{k=1}^{\infty} a_k \text{ converges}} \quad \Leftrightarrow \quad \boxed{\sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges}}$$

Proof. Let $\left(s_n\right)$ and $\left(t_m\right)$ be sequences such that

$$s_n = \sum_{r=1}^{n} a_r \text{ and } t_m = \sum_{r=0}^{m} 2^r a_{2^r} .$$

In other words,

$$t_m = \sum_{r=0}^{m} 2^r a_{2^r} = a_1 + 2a_2 + 4a_4 + 8a_8 + \ldots + 2^m a_{2^m} .$$

Given

Sequence (a_k) is decreasing

$a_k \geq 0$ for all $k \in \mathbb{N}$

Since $a_k \geq 0 \ \forall k \in \mathbb{N}$, it follows that for all $n, m+1 \in \mathbb{N}$:

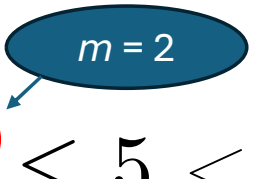
$$s_{n+1} - s_n = a_{n+1} \geq 0 \text{ and } t_{m+1} - t_m = 2^{m+1} a_{2^{m+1}} \geq 0 .$$

Thus, both $\left(s_n\right)$ and $\left(t_m\right)$ are monotone increasing.

What is their relation?

$$\left(s_n\right) \text{ and } \left(t_m\right)$$

We help you to see the pattern...



If $n = 5$, then $2^{2} \leq 5 < 2^{3} = 2^{2+1}$. Take $m = 2$.

$$s_5 = a_1 + a_2 + a_3 + a_4 + a_5$$

$$t_2 = a_1 + 2a_2 + 4a_4 = a_1 + a_2 + a_2 + a_4 + a_4 + a_4 + a_4$$

$$2s_5 = 2a_1 + 2a_2 + 2a_3 + 2a_4 + 2a_5$$

Note: (a_k) is decreasing

Can you see that $s_5 \leq t_2 \leq 2s_5$?

Try when $n = 3$, then $2^1 \leq 3 < 2^{1+1}$ (here $m = 1$).

Can you get $s_3 \leq t_1 \leq 2s_3$?

What about $n = 6$, what is the m?

What is the general pattern we observe?

$$\boxed{s_n \leq t_m \leq 2s_n}$$

Can we prove this is correct?

Prove that $\forall n \in \mathbb{N}\ \exists m \in \mathbb{N} \ni s_n \leq t_m \leq 2s_n$.

Note that $\forall n \in \mathbb{N}\ \exists m \in \mathbb{N} \ni 2^m \leq n < 2^{m+1}$.

We consider two cases below:

Case 1: When $n < 2^{m+1}$. Case 2: When $2^m \leq n$.

We need to satisfy both cases.

Case 1: If $n < 2^{m+1}$.

$$
\begin{aligned}
s_n &= a_1 + a_2 + a_3 + \cdots + a_n \\
&\leq a_1 + \left(a_2 + a_3\right) + \left(a_4 + a_5 + a_6 + a_7\right) + \cdots + \left(a_{2^m} + a_{2^m+1} + \cdots + a_{2^{m+1}-1}\right) \\
&\leq a_1 + \left(a_2 + a_2\right) + \left(a_4 + a_4 + a_4 + a_4\right) + \cdots + \left(a_{2^m} + a_{2^m} + \cdots + a_{2^m}\right) \\
&= a_1 + 2a_2 + 2^2 a_4 + \cdots + 2^m a_{2^m} = t_m
\end{aligned}
$$

add these terms on purpose ↓↓

$\because a_2 \geq a_3 \geq a_4 \geq a_5 \geq a_6 \geq \cdots$

Case 2: If $2^m \le n$.

On the other hand, since $2^m \le n$, it follows that

$$
\begin{aligned}
t_m &= a_1 + 2a_2 + 2^2 a_4 + 2^3 a_8 + \cdots + 2^m a_{2^m} \\
&= 2\left(\tfrac{1}{2}a_1 + a_2 + 2a_4 + 2^2 a_8 + \cdots + 2^{m-1} a_{2^m}\right) \\
&\le 2\Bigg(a_1 + a_2 + (a_4 + a_4) + (a_8 + a_8 + a_8 + a_8) + \cdots + (\overbrace{a_{2^m} + \cdots + a_{2^m}}^{2^{m-1}\text{ times}})\Bigg) + 2a_{2^m+1} + \cdots + 2a_n \\
&\le 2\left(a_1 + a_2 + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots + (a_{2^{m-1}+1} + \ldots + a_{2^m}) + a_{2^m+1} + \cdots + a_n\right) \\
&= 2\left(a_1 + a_2 + \cdots + a_n\right) = 2s_n \qquad\qquad \because a_3 \ge a_4 \ge a_5 \ge a_6 \ge \cdots
\end{aligned}
$$

add these terms on purpose
↓↓

Thus, $\forall n \in \mathbb{N}\ \exists m \in \mathbb{N} \ni s_n \leq t_m \leq 2s_n$.

Particularly, $s_n \leq t_m$ and $t_m \leq 2s_n$.

Since $\left(s_n\right)$ and $\left(t_m\right)$ are monotone increasing,

it follows by MCT that (why not mention bounded?)

$\left(s_n\right)$ is convergent if and only if $\left(t_m\right)$ is convergent.

(Note that a sequence converges implies that it is bounded.)

In other words,

$\left(s_n\right)$ converges $\Rightarrow$ $\left(2s_n\right)$ converges $\Rightarrow$ $\left(2s_n\right)$ bounded

$\Rightarrow$ $\left(t_m\right)$ increase and bounded $\Rightarrow$ $\left(t_m\right)$ converges by MCT

$\left(t_m\right)$ converges $\Rightarrow$ $\left(t_m\right)$ bounded

$\Rightarrow$ $\left(s_n\right)$ increase and bounded $\Rightarrow$ $\left(s_n\right)$ converges by MCT

Hence, $\left(s_n\right)$ converges $\Leftrightarrow$ $\left(t_m\right)$ converges.

Application of Theorem 4.5



Theorem 4.5 is used to prove Theorem 4.6 (p-series test)



We next prove Theorem 4.6.

Theorem 4.6 (p-series test)

The series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ is convergent if and only if $p > 1$.

Proof. If $p < 0$, then $\left(\frac{1}{n^p}\right) \to \infty$. If $p = 0$, then $\left(\frac{1}{n^p}\right) \to 1$.

Thus, when $p \leq 0$, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges (by divergence test).

When $p > 0$, note that $\left(\frac{1}{n^p}\right)$ is decreasing and $\frac{1}{n^p} \geq 0 \ \forall n \in \mathbb{N}$.

Observe this

for $p > 0$

If $a_k = \frac{1}{k^p}$. Then we have

$$s_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n} \frac{1}{k^p} \text{ and}$$

$$t_m = \sum_{k=0}^{m} 2^k a_{2^k} = \sum_{k=0}^{m} \left(2^k \cdot \frac{1}{2^{kp}}\right) = \sum_{k=0}^{m} \left(\frac{2^k}{2^{kp}}\right) = \sum_{k=0}^{m} \left(\frac{1}{2^{k(p-1)}}\right).$$

By Theorem 4.4 it follows that

$$\sum_{k=1}^{\infty}\frac{1}{k^p} \text{ converges if and only if } \sum_{k=0}^{\infty}\frac{2^k}{\left(2^k\right)^p}=\sum_{k=0}^{\infty}\frac{1}{\left(2^k\right)^{p-1}} \text{ converges.}$$

$$2^{k(p-1)}=\left(2^{(p-1)}\right)^k$$

Now, this is a geometric series $\sum_{n=1}^{\infty}x^n$ with $x=\frac{1}{2^{p-1}}$, and it is convergent

if and only if $\left|x\right|=\left|\frac{1}{2^{p-1}}\right|=\frac{1}{2^{p-1}}<1\Leftrightarrow p>1\,.$

Recall: Definition of absolute and conditional convergence

(a) A series $\sum_{k=1}^{\infty} a_k$ is said to be absolutely convergent

if $\sum_{k=1}^{\infty} \left|a_k\right|$ is a convergent series.

(b) If $\sum_{k=1}^{\infty} a_k$ does not converge absolutely, i.e., $\sum_{k=1}^{\infty} \left|a_k\right|$ is divergent;

but $\sum_{k=1}^{\infty} a_k$ converge, then it is said to be conditionally convergent.

Recall: Theorem 3.5 (Series properties)

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be convergence series that converge to L and M, respectively.

Then $\sum_{k=1}^{\infty} \alpha a_k$ and $\sum_{k=1}^{\infty} \left(\alpha a_k + \beta b_k\right)$ are convergence series whose converge to αL and $\alpha L + \beta M$, respectively.

Theorem 4.12. Absolute Convergence

If a series $\sum_{k=1}^{\infty} a_k$ converges absolutely, then it converges.

Proof. Note that $0 \le a_k + |a_k| \le 2|a_k| \quad \forall k \in \mathbb{N}$.

$$\sum_{k=1}^{\infty} |a_k| \text{ coverges} \xRightarrow[\text{(comparison test)}]{} \sum_{k=1}^{\infty} \left(a_k + |a_k|\right) \text{ converges}$$

$$\xRightarrow[\text{(Theorem 3.5)}]{} \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \left(a_k + |a_k|\right) - \sum_{k=1}^{\infty} |a_k| \text{ converges}.$$

Theorem 4.16. Ratio Test

Let $\sum_{k=1}^{\infty} a_k$ be a series with $a_k > 0 \ \forall k \in \mathbb{N}$.

Define $L = \lim\left(\frac{a_{k+1}}{a_k}\right)$. Then $L \geq 0$. Moreover,

(i) if $L < 1$ then $\sum_{k=1}^{\infty} a_k$ converges; and

(ii) if $L > 1$ then $\sum_{k=1}^{\infty} a_k$ diverges.

The test is inconclusive if $L = 1$.

We prove for the case without the absolute sign because we can apply the absolute convergence theorem when necessary!

Proof. Since $a_k > 0 \ \forall k \in \mathbb{N}$, it follows that $\frac{a_{k+1}}{a_k} > 0$.

Recall Theorem A in Chapter 2:

If $\alpha \ \boxed{\leq or <} \ x_k \boxed{\leq or <} \ \beta \ \forall k \in \mathbb{N}$, then $\alpha \leq \lim\left(x_k\right) \leq \beta$.

Let $\left(\frac{a_{k+1}}{a_k}\right)$ be a sequence and $L = \lim\left(\frac{a_{k+1}}{a_k}\right) \geq 0$.

$\mathbb{R}$ is dense

Case 1. Suppose $L < 1$. $\exists \alpha \in \mathbb{R}^+ \ni 0 \leq L < \alpha < 1$.

Choose $\varepsilon = \alpha - L > 0$. Then $\exists N \in \mathbb{N}$ such that when $k > N$ implies that

$$\left| \frac{a_{k+1}}{a_k} - L \right| < \varepsilon = \alpha - L$$

$$\Rightarrow L - \alpha + L < \frac{a_{k+1}}{a_k} < L + \alpha - L$$

$$\Rightarrow 2L - \alpha < \boxed{\frac{a_{k+1}}{a_k} < \alpha}$$

$$\Rightarrow a_{k+1} < \alpha a_k$$

Note that from induction, when $k = N+1, N+2, \ldots$

$$a_{N+2} < \alpha a_{N+1} \Rightarrow a_{N+3} < \alpha a_{N+2} < \alpha^2 a_{N+1}$$
$$\Rightarrow a_{N+4} < \alpha a_{N+3} < \alpha^2 a_{N+2} < \alpha^3 a_{N+1} \ldots \quad .$$
$$etc$$

So in general,

$$a_{N+1+r} < \alpha^r a_{N+1} \ \forall r \in \mathbb{N} \, .$$

Since $\sum_{r=1}^{\infty}\left(\alpha^r a_{N+1}\right) = a_{N+1}\sum_{r=1}^{\infty}\alpha^r$ converges

$(0 < \alpha < 1$ by geometric series),

$$\Rightarrow \sum_{r=1}^{\infty} a_{N+1+r} = \sum_{k=N+2}^{\infty} a_k \text{ converges}$$

$$\Rightarrow \sum_{k=1}^{\infty} a_k \text{ converges (by comparison test).}$$

Case 2. Suppose $L > 1$. $\exists \alpha \in \mathbb{R}^+ \ni 1 < \alpha < L$.

Choose $\varepsilon = L - \alpha > 0$. Then $\exists N \in \mathbb{N}$ such that when $k > N$ implies that

$$\left| \frac{a_{k+1}}{a_k} - L \right| < \varepsilon = L - \alpha$$

$$\Rightarrow -L + \alpha + L < \frac{a_{k+1}}{a_k} < L - \alpha + L$$

$$\Rightarrow \boxed{\alpha < \frac{a_{k+1}}{a_k}} < 2L - \alpha$$

$$\Rightarrow \alpha a_k < a_{k+1}$$

Note that from induction, when $k = N+1, N+2, \ldots$

$$\alpha a_{N+1} < a_{N+2} \Rightarrow \alpha^2 a_{N+1} < \alpha a_{N+2} < a_{N+3}\,.$$

So in general,

$$\alpha^r a_{N+1} < a_{N+1+r} \;\; \forall r \in \mathbb{N}\,.$$

Since $\sum_{r=1}^{\infty}\left(\alpha^r a_{N+1}\right) = a_{N+1}\sum_{r=1}^{\infty}\alpha^r$ diverges

$(\alpha > 1$ by geometric series),

$$\Rightarrow \sum_{r=1}^{\infty} a_{N+1+r} = \sum_{k=N+2}^{\infty} a_k \text{ diverges}$$

$$\Rightarrow \sum_{k=1}^{\infty} a_k \text{ diverges (by comparison test).}$$

When $L = \infty$, the proof is similar to the case of $L > 1$.

$$\text{Suppose } L = \infty.$$

$$\forall M \in \mathbb{R}^+ \; \exists N \in \mathbb{N} \ni k > N$$

$$\Rightarrow \frac{a_{k+1}}{a_k} > M \; \forall k \in \mathbb{N}.$$

$$(\text{In particular, } M > 1 \text{ is chosen.})$$

Thus, $a_{k+1} > Ma_k \ \forall k > N$

$$\Rightarrow a_{N+2} > Ma_{N+1}$$

$$\Rightarrow a_{N+3} > Ma_{N+2} > M^2 a_{N+1}$$

$$\vdots$$

$$\Rightarrow a_{N+1+r} > M^r a_{N+1} \ \forall r \in \mathbb{N}$$

Since $\sum_{r=1}^{\infty}\left(M^r a_{N+1}\right) = a_{N+1}\sum_{r=1}^{\infty} M^r$ diverges,

it follows that $\sum_{r=1}^{\infty} a_{N+1+r} = \sum_{k=N+2}^{\infty} a_k$ diverges

and so $\sum_{k=1}^{\infty} a_k$ diverges.

Case 3. We consider the following series whose give $L = 1$.

$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \text{ diverges but } \sum_{k=1}^{\infty} \frac{1}{k^2} \text{ converges.}$$

Overview

We prove Theorem 4.9 (Alternating series test).

To prove it, we need Q15a), Worksheet 4.

We apply Cauchy criterion (for sequence)

Question 15.

(a) Let $\left(a_k\right)$ be a decreasing sequence with $\lim\left(a_k\right) = 0$.

Prove that $a_k \geq 0 \ \forall k \in \mathbb{N}$.

Proof. (a) Suppose $\exists m \in \mathbb{N} \ni a_m < 0$.

Given that $\lim\left(a_k\right) = 0$. Thus,

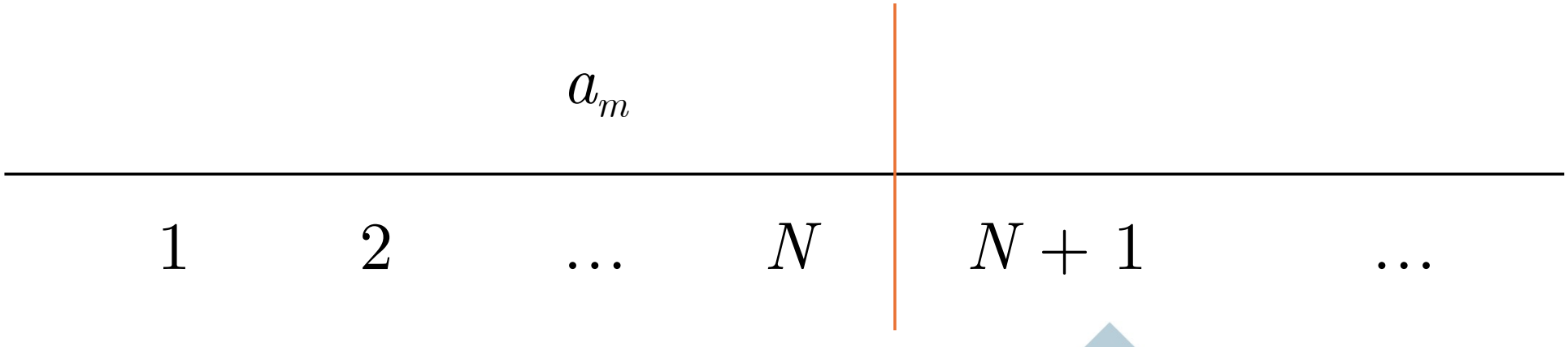
$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\left|a_k\right| < \varepsilon \ \forall k > N$.

Let $\varepsilon = -\frac{a_m}{2} > 0$. Then $\frac{a_m}{2} < a_k < -\frac{a_m}{2} \Rightarrow m \leq N$.

If $m > N$, then pick $n = m$ which implies that

$a_m > \frac{a_m}{2} \Rightarrow 1 < \frac{1}{2}$, a contradiction.

In other words, $a_m < 0$ exists for finitely many times and $m \leq N$.



Given that the sequence is decreasing,

and since $a_m < 0$, it follows that

$$0 > a_m \geq a_{m+1} \geq \cdots \geq a_N \geq a_{N+1} \geq \cdots,$$

which means $a_{N+1} < 0$, a contradiction.

Theorem 4.9 (Alternating series test)

Let $\left(a_k\right)$ be a monotone decreasing sequence such that $\lim\left(a_k\right) = 0$.

Then $\sum_{k=1}^{\infty}(-1)^k a_k$ is convergent.

Proof.(We attack the problem using the Cauchy criterion)

For $n > m > N$, let $s_n = \sum_{k=1}^{n} (-1)^k a_k$ and $s_m = \sum_{k=1}^{m} (-1)^k a_k$. Then

$$\begin{aligned} \left| s_n - s_m \right| &= \left| (-1)^{m+1} a_{m+1} + (-1)^{m+2} a_{m+2} + (-1)^{m+3} a_{m+3} + \cdots + (-1)^n a_n \right| \\ &= \left| a_{m+1} - a_{m+2} + a_{m+3} - \cdots + (-1)^{n-m-1} a_n \right| \end{aligned}$$

$\left(a_k\right)$ is decreasing and $\lim\left(a_k\right) = 0$, it follows that $a_k \geq 0 \ \forall k \in \mathbb{N}$.

(Q15a) just now!)

Next, we claim that $\left|s_n - s_m\right| \leq \left|a_{m+1}\right| \; \forall n > m$.

If $(-1)^{n-m-1} a_n = -a_n$, then by the fact that $\left(a_n\right)$ is monotone decreasing,

$$\begin{aligned}
\left|s_n - s_m\right| &= \left|a_{m+1} - a_{m+2} + a_{m+3} - a_{m+4} + a_{m+5} \cdots - a_{n-2} + a_{n-1} - a_n\right| . \\
&\leq \left|a_{m+1} - a_{m+2} + a_{m+2} - a_{m+4} + a_{m+4} \cdots - a_{n-2} + a_{n-2} - a_n\right| \\
&= \left|a_{m+1} - a_n\right| \\
&< \left|a_{m+1}\right|
\end{aligned}$$

If $(-1)^{n-m-1}a_n = a_n$, then by the fact that $\left(a_n\right)$ is monotone decreasing,

$$\begin{aligned}
\left|s_n - s_m\right| &= \left|a_{m+1} - a_{m+2} + a_{m+3} - \cdots - a_{n-1} + a_n\right| \\
&\leq \left|a_{m+1} - a_{m+2} + a_{m+2} - \cdots - a_{n-1} + a_{n-1}\right| \\
&= \left|a_{m+1}\right|
\end{aligned}$$

As $\lim\left(a_k\right) = 0$, it follows that $\forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|a_m\right| < \varepsilon \ \forall m > N$.

$$\left|s_n - s_m\right| \leq \left|a_{m+1}\right| < \varepsilon \text{ for all } n > m > N.$$

Hence $\left(s_n\right)$ is Cauchy and thus $\sum_{k=1}^{\infty} (-1)^k a_k$ converges.

Application of Cauchy criterions

- We stress that both the Cauchy criterions are EXTREMELY useful (more than what we had apply on proving several theorems earlier).
- Try Worksheet 4: Q8, Q15b), Q16a), Q17.